THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 2: **Power Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $5^x \times 5^2 = 5^9$, then x is: 1

(a) 7 **(b)** 9 **(c)** 18 **(d)** 11

2. If $(5^4)^x = 5^{20}$, then x is: 1

(a) 80 **(b)** 5 **(c)** 24 **(d)** 16

3. $4^2 \times 25^2$ equals: 1

(a) 10000 **(b)** 2500 **(c)** 641 **(d)** 200

4. The product $(1.5 \times 10^2) \times (4 \times 10^5)$ in scientific notation is: 1

(a) 6×10^7 **(b)** 6×10^8 **(c)** 6×10^6 **(d)** 60×10^7

5. The value of $3^4 - 4^3$ is: 1

(a) 0 **(b)** -17 **(c)** 17 **(d)** 145

6. $3^{-1} + 4^{-1}$ equals: 1

(a) $\frac{2}{7}$ **(b)** $\frac{1}{7}$ **(c)** $\frac{7}{12}$ **(d)** $\frac{1}{12}$

7. $\dfrac{2^9 \times 2^5}{2^7}$ equals: 1

(a) 2^7 **(b)** 2^{11} **(c)** 2^{38} **(d)** 2^{21}

8. The value of $\left(\frac{4}{3}\right)^{-3}$ is: 1

(a) $\frac{27}{64}$ **(b)** $\frac{9}{4}$ **(c)** $-\frac{27}{64}$ **(d)** $\frac{64}{27}$

9. If $2^x \times 2^6 = 2^{15}$, then x is: 1

(a) 21 **(b)** 11 **(c)** 9 **(d)** 90

10. If $(3^5)^x = 3^{35}$, then x is: 1

(a) 30 **(b)** 7 **(c)** 40 **(d)** 175

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find x if $3^x = \dfrac{1}{27}$. 2

12. Write $(9 \times 10^{-2}) \div (3 \times 10^2)$ in scientific notation. 2

13. Find the value of $(5^2)^3 \div 5^4$. 2

14. Find x if $5^x = \frac{1}{25}$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Simplify $\frac{5^4 \times 3^5}{5^2 \times 3^3}$ and find its value. 3

16. Evaluate $3^{-2} + 2^{-5}$ as a single fraction. 3

17. A factory makes 4×10^5 bolts each day. How many bolts does it make in 30 days? Give the answer in scientific notation. 3

OR

Simplify $\frac{2^8 \times 7^3}{2^5 \times 7^1}$ and find its value. 3

18. Evaluate $4^{-2} + 5^{-2}$ as a single fraction. 3

OR

A factory makes 6.4×10^2 bolts each day. How many bolts does it make in 25 days? Give the answer in scientific notation. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Simplify $\frac{5^1 \times 2^{-2}}{5^{-2} \times 2^1}$ and write the answer with positive exponents. Then find its value. 5

OR

The diameter of a cell is 5×10^{-4} cm. How many such cells placed side by side in a straight line make a length of 15×10^1 cm? Give the answer in scientific notation. 5

20. Simplify $\frac{2^4 \times 5^{-1}}{2^{-1} \times 5^1}$ and write the answer with positive exponents. Then find its value. 5

OR

The diameter of a cell is 2.5×10^{-3} cm. How many such cells placed side by side in a straight line make a length of 7.5×10^2 cm? Give the answer in scientific notation. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND